AMRITA VISHWA VIDYAPEETHAM
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COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

Team ID: *[GEN26-A0H-I018]*

Team Name: *[Secret Avengers]*

Domain: *[1]*

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Presentation Outline

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Problem Statement

Buildings such as college classrooms, laboratories and offices are consuming electricity unnecessarily when energy usage does not match actual occupancy and demand. Existing approaches, including manual monitoring and fixed schedules, may not respond effectively to changing weather, schedules and building activity. Our project aims to develop an AI-powered system that predicts electricity demand, identifies potential energy wastage, and recommends adaptive energy-saving actions while maintaining comfort and safety.

Problem Understanding

- Who gets affected by it?
  - College and university facility managers responsible for electricity consumption.
  - Students and faculty using classrooms, laboratories and libraries.
  - Office and commercial building operators managing energy costs.
  - Individuals who have to pay current bill for residential purposes
- What causes the problem?
  - Fixed schedules may not adapt to unexpected changes in occupancy.
  - Monitoring energy only when it occurs.
  - Weather, academic schedules and equipment usage cause electricity demand to fluctuate.
  - Energy consumption data and building schedules may not be analysed together to anticipate demand.
- How does it affect users or stakeholders?
  - Facility managers: Spend more time monitoring usage and investigating unusual consumption.
  - Institutions and businesses: May incur avoidable electricity costs.

Existing Gaps & Challenges

- Existing gap or limitation 1
  Manual energy monitoring: Time-consuming and dependent on staff noticing unusual consumption.

- Existing gap or limitation 2
  Fixed operating schedules: May not adapt to unexpected changes in occupancy, weather or activities.

- Main challenge that remains unsolved
  Motion and occupancy sensors: Can respond to presence, but may not independently predict future demand or account for every type of equipment.
  The gap we aim to address: integrating demand prediction, potential-waste detection and context-aware energy-saving recommendations into one software-based workflow.

Our Proposed Solution

- Core idea or product:
  An AI-powered predictive and adaptive energy optimisation system for smart buildings that forecasts electricity demand, identifies potential energy wastage and recommends suitable energy-saving actions.

- How the solution works at a high level
  The system analyses historical electricity consumption along with relevant factors such as weather, time, occupancy and academic schedules. A machine-learning model predicts future energy demand, identifies unusual consumption patterns and generates recommendations to improve energy efficiency while maintaining occupant comfort and safety.

- Main value delivered to users
  The system helps campus administrators and facility managers make data-driven energy-management decisions, reduce unnecessary electricity consumption, identify potential savings and minimise dependence on manual monitoring and fixed operating schedules.

Technical Approach

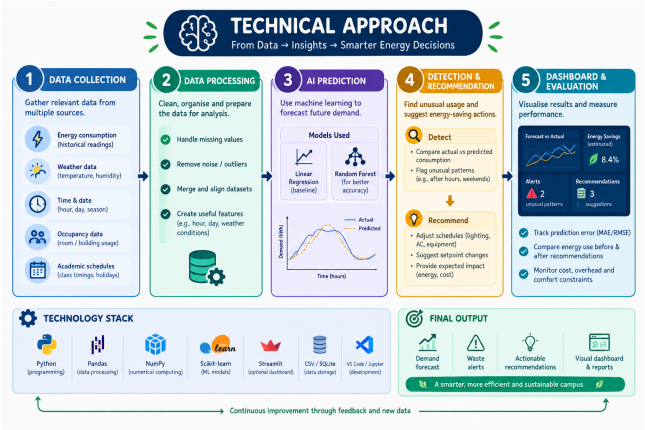


Figure: Technical approach diagram

[Data is collected, then cleaned with Pandas/NumPy, then Scikit-learn analyses it, then converted into alerts and recommendations, then displayed through Streamlit and then it is evaluated to improve future predictions..]

Tech Stack

Frontend: [Displays electricity demand forecasts, unusual-usage alerts, and energy-saving

Database: [CSV / SQLite: Stores historical energy data, weather data, occupancy information, and model results.]

Innovation & Uniqueness

- What is new or different?
  Our solution combines electricity demand forecasting, unusual-usage detection, and context-aware energy-saving recommendations in one system. Instead of relying only on fixed schedules or monitoring current consumption, it uses historical data and factors such as weather, occupancy, time, and academic schedules to support more adaptive decisions.

- Why is this approach suited to the problem?
  Building electricity consumption changes with occupancy, weather, class timings, and equipment usage. AI-based forecasting can identify expected demand, while comparing predicted and actual consumption can help flag unusual patterns. This makes the approach suitable for helping facility managers make informed decisions without requiring expensive hardware in the initial prototype.

- What is the key differentiator?
  integration of prediction, waste detection, and actionable recommendations in a single, software-first workflow

Feasibility & Scalability

- **Feasibility:** Build a software prototype using Python, Pandas, NumPy, Scikit-learn, and CSV/SQLite. Begin with public or simulated data, train forecasting models, generate recommendations, and display results through an optional Streamlit dashboard

- **Scalability:** Start with one classroom or laboratory and expand to multiple rooms, buildings, and campuses by integrating additional datasets and validating the models for each setting.

- **Risks:** Limited or inaccurate data may reduce prediction accuracy; simulated data may not represent real building behaviour; privacy concerns may arise from occupancy information; recommendations may affect comfort if poorly designed. Mitigate these risks through data validation, privacy-conscious data handling, model evaluation, and human review of recommendations.

Cost for Product (Optional)

- Development or hardware cost: 0 additional cost for the initial software prototype, assuming existing laptops and internet access are available. No dedicated hardware is required initially.
- Hosting or maintenance cost: 0 initially using local execution and open-source tools; future cloud hosting and maintenance may incur additional costs.
- Estimated total:0 additional software cost for the initial prototype, excluding labour, electricity, and any optional hardware or cloud services.

Use Cases (compulsorily 3 use cases must be there)

- Use case 1: College classrooms — predict electricity demand based on class timings, occupancy, and weather to identify potential energy waste.
- Use case 2: Computer laboratories — analyse historical consumption and operating schedules to flag unusual electricity usage and recommend suitable adjustments.
- Use case 3: Libraries and study spaces — forecast demand during different hours and recommend energy-saving actions while maintaining lighting, cooling, and user comfort.
  Note: Additional slides may be included only for this section.

Impacts and Benefits

- Benefit to the target users: Helps campus administrators and facility managers forecast electricity demand, identify potential inefficiencies, and make data-driven energy-management decisions.
- Social, economic, or environmental impact: Supports responsible electricity consumption, potential reductions in energy expenditure, and lower emissions associated with electricity generation.
- Expected measurable outcome: Evaluate forecasting accuracy using MAE and RMSE; measure changes in electricity consumption and cost after recommendations only when reliable before-and-after data are available.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 7	Target 7.3	Affordable and Clean Energy	Supports improvements in energy efficiency through demand forecasting and energy-saving recommendations.
SDG 12	Target 12.2	Responsible Consumption and Production	Encourages more efficient use of electricity and helps identify potential waste.

Individual Member Contribution

Member	Contribution or responsibility
Sree Gugan V S	pre ideathon ppt, all registrations, Team Leader
T Harshil	research work, idea generation, chart making
Theruvai Nithesh	research work, idea generation, chart making
Yenigalla Akshai	research work, idea generation, chart making

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